Year 13 Further Maths - Series

Similar questions - Method of differences with products and pi^2/6

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning. It is given that $\sum_{r=k}^{48} \frac{1}{r(r+1)(r+2)} = \frac{43}{3675}$ for some positive integer k. Determine the value of k. **[7]**

2

(a) Show that $\sum_{r=1}^{n} \frac{3r+2}{r^2(r+1)} = 1 - \frac{1}{n+1} + a\sum_{r=1}^{n} \frac{1}{r^2}$ where a is an integer to be determined. **[6]**

(b) You are given that $\sum_{r=1}^{\infty} \frac{1}{r^2} = \frac{\pi^2}{6}$. Show that $\sum_{r=1}^{\infty} \frac{3r+2}{r^2(r+1)} = 1 + \frac{\pi^2}{3}$. **[2]**

Question 1

$\dfrac{1}{r(r+1)(r+2)} = \dfrac{1}{2}\left[\dfrac{1}{r(r+1)} - \dfrac{1}{(r+1)(r+2)}\right]$. The sum telescopes: $\dfrac{1}{2}\left[\dfrac{1}{k(k+1)} - \dfrac{1}{49\times 50}\right] = \dfrac{43}{3675}$. So $\dfrac{1}{k(k+1)} = \dfrac{86}{3675} + \dfrac{1}{2450} = \dfrac{172+3}{7350} = \dfrac{175}{7350} = \dfrac{1}{42}$. Then $k(k+1) = 42$, $k = 6$

Question 2

(a) $\dfrac{3r+2}{r^2(r+1)} = \dfrac{2}{r^2} + \dfrac{1}{r} - \dfrac{1}{r+1}$ (partial fractions). Summing, $\sum\left(\dfrac{1}{r} - \dfrac{1}{r+1}\right) = 1 - \dfrac{1}{n+1}$, so the sum is $1 - \dfrac{1}{n+1} + 2\sum\dfrac{1}{r^2}$, giving $a = 2$

(b) As $n \to \infty$: $1 - 0 + 2\times\dfrac{\pi^2}{6} = 1 + \dfrac{\pi^2}{3}$